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Business Case

AUTOMATED INVOICE PROCESSING

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1. Executive summary

Northwind Enterprise's accounts payable department is having a hard time processing invoices and paying suppliers on-time despite having 8 clerks and 2 managers overseeing the activity. With over 4000 invoices monthly, and that number increases during peak business periods, the staff are having a hard time keeping up with demand and the situation is worsening. The situation is creating a large number of errors, causing delays in payment and dissatisfied suppliers. It is creating low morale between the workers in the department and communication with other departments is difficult because of the lack of information of where invoices are in the invoice processing cycle.

An investment for an automated invoice processing system using OCR (Optical Character Recognition) has been presented as a solution to help solve the problems in the accounts payables (AP) department. The fully automated invoice processing program will fully integrate into the current ERP. The new program will be able to alleviate all the pain points for the department and meet many of the business objectives put forth by the company.

- When fully integrated, OCR will be able to read, scan, and enter new invoices into Northwind's ERP system. This will reduce the high error rates, remove the manual invoice processing by clerks and prevent overworking.
- Provide real-time status of invoices, through the processing cycle so supplier questions of payment can be answered accurately and timely.
- Because processing will mostly be automated, processing costs will decrease, processing time will decrease, and scalability will be achievable during peak business periods.

The initial investment is \$190,000 with a monthly licencing fee of \$5000 per month (\$60,000 per year). Within the first year, Northwind will see a savings of \$230,000. Which means the company will be able to recoup initial investment within the first year. Within 4 years, Northwind should see a 384.21% Return on Investment (ROI) or an estimated \$730,000 accumulated savings.

The automatic processing system is the most viable solution for the company and its ability to scale means that as the business grows, the processing system is able to accommodate the growth. In addition, Northwind should see a relative quick return on investment by implementing an automatic processing system. Lastly, the automatic processing system will be able to alleviate many of the present pain points with the present manual invoice processing. As well, the automatic processing system will be able to meet many of the business objectives set by the company.

2. Introduction

Northwind Enterprises, a mid-sized company specializing in wholesale food and beverage distribution, relies heavily on its Accounts Payable (AP) department. With over 500 suppliers and a monthly volume of approximately 4,000 invoices, the AP department's role is pivotal to the company's operations. The department, consisting of eight clerks and two managers, is responsible for the end-to-end invoice processing process.

Northwind Enterprises currently processes invoices manually, which leads to high error rates, delays, and increased labor costs.

Northwind Enterprises uses an ERP system that manages financial data, but the system is not integrated with manual invoice processing. The company has a robust IT infrastructure capable of supporting new software implementations.

Current Accounts Payable process

The current Accounts Payable (AP) process at Northwind Enterprises begins with the clerks receiving a paper invoice. They then manually enter this invoice into the accounting system, which is part of the ERP. The clerks validate the invoice data against the purchase orders, checking vendor details, quantities ordered, and values. They also need to ensure that the goods or services invoiced have actually been delivered.

Once validated, the paper invoice is sent to the relevant manager for approval. Once approved, the original invoice is returned to the AP clerk, who sends it to the manager for payment. Once the payment is processed and recorded, the AP clerk receives the invoice back. He then scans it and files it (physically and digitally).

Current Pain Points:

- High error rates lead to payment delays and supplier dissatisfaction.
- The time-consuming process causes clerks to be overworked.
- The lack of real-time tracking and visibility into invoice status adds to supplier dissatisfaction, as nobody can answer their questions about when the invoice will be paid.
- There are frequent miscommunications between departments (finance, procurement).
- It is difficult to scale the process during peak business periods.

3. Business Objectives

The Northwind Enterprises management agreed that the proposed invoicing solution must deliver on the following business objectives:

- Reduce invoice processing errors by 90%
- Decrease the average invoice processing time by 50% within the first year.
- Lower the overall invoice processing costs.
- Provide the status of invoices in real-time throughout the processing cycle.
- Achieve a 95% on-time payment rate and reduce supplier complaints about payment issues by 80% within the first year.
- Develop a scalable invoice processing system to handle increased volumes during peak business periods without additional staffing.

4. The Initiatives or Solutions Considered

Northwind Enterprises are considering the following 3 solutions

1. Keeping the manual process and address the issue through process changes and more strict quality monitoring
 - a. Pros:
 - i. low to no cost to business upfront
 - ii. will not require a development team
 - iii. staff and business analyst can optimize existing process
 - b. Cons:
 - i. Unable to address increasing invoice volume
 - ii. Process optimization has limitations
 - iii. Manual entry will still cause errors. Invoices will need a quality control step added.
 - iv. Company will still struggle to pay vendors on-time. Over time this will be a competitive disadvantage.
 - v. Eventually, an alternate solution will have to be looked at when invoice volumes grow.
2. Hire more staff.
 - a. Pros:
 - i. Will ease the workload for existing accounts payable clerks.

- ii. Easy way to address invoice volume. As invoice volume increase, another clerk can be added.
 - b. Cons:
 - i. Expensive to hire employee when automation can handle volume faster and more reliably
 - ii. Number of employees will reach a capacity. Will there be room for additional staff?
 - iii. Adding more staff only temporarily fixes the problem, long processing times and manual-data-entry-errors will still persist and need to be addressed.
3. Invest in fully automated invoice processing system.
- a. Pros:
 - i. Able to enter invoice and invoice validation. Can integrate to ERP system.
 - ii. Will not require additional staff.
 - iii. Approvals can also be automated
 - iv. OCR scan will remove errors due to data entry. This will reduce processing time and allow vendors to be paid on-time.
 - b. Cons:
 - i. Expensive upfront costs
 - ii. Licencing fee is \$5000 a month
 - iii. Need to find and hire a development team
 - iv. Trust the system will deliver the requirements they requested

5. Capability Assessment

Technical Infrastructure

Northwinds Enterprise's ERP system does not integrate with the manual invoice processing. Due to manual invoice entry, the system has many mistakes. But Northwind has a robust ERP system and is able to support new software implementation. Automation implementation can happen on the existing system without further investments in infrastructure. OCR technology will need software and hardware to integrate into ERP system. Data integrity, data migration, data security, and seamless integration to current ERP system will still need to be facilitated with the IT team.

Human Resources

The current AP department has 8 clerks and 2 managers with experience handling manual invoice processing and an IT team with technical expertise. Transitioning into an automated process will require training for existing AP staff so they can navigate the new system efficiently. Resistance to change, and helping staff transition into new roles will have to be addressed and implemented to help the transition into a new automated invoice processing system.

Financial Resources

Northwinds has budgeted \$250,000 to implement the automated invoice processing. (\$190,000 for software implementation and \$5,000 per month for licensing fees). Planning for unexpected costs like delays in implementation, or difficulty in data migration may help prevent any budget over run. This may mean adjusting the expectation of the implementation or having a contingency fund for unexpected cost to this project.

Organizational Readiness

The present system is prone to errors and the end-to-end processing of invoices is long and arduous. The organization has budgeted the funds to make the change to an automated invoice process system. Support from higher-up for this project is there. The organization will also develop new procedures for invoice processing under the new system. New processes will have to optimize the features of invoice automation and ensure workflow is efficient and effective.

6. Financial Analysis

The financial analysis is based on the following figures and assumptions:

- Current staff complement 8 Accounts payable clerks and 2 managers.
- Annual wage for an AP clerk is \$50,000. Annual wage for a manager is \$90,000.
- The software integration and customization will cost \$190,000 once off and \$5,000 per month in licencing fees.
- It is expected that with the new system the company will be able to reduce AP staff to 4 clerks and 1 manager.

Detailed Analysis

- Current Annual Labor Cost: \$580,000 (8 clerks x \$50,000 + 2 managers x \$90,000 = \$580,000)
- Future Annual Labor Cost: \$290,000 (4 clerks x \$50,000 + 1 manager x \$90,000 = \$290,000)
- Annual Labor Savings: \$290,000 (\$580,000 - \$290,000 = \$290,000)
- Annual Licensing Cost: \$60,000 (\$5,000 x 12 months)
- Net Annual Savings: \$230,000 (\$290,000 - \$60,000 = \$230,000)
- Return on Investment (ROI) over 4 years (%): 384.21%
- The ROI is calculated as follows $((\$230,000 \times 4 \text{ years}) - \$190,000) / \$190,000 \times 100\% = 384.21\%$
- Payback Period (years): 0.83 years

Summary of Net Benefit by Year

- Year 1: \$40,000
- Year 2: \$270,000
- Year 3: \$500,000
- Year 4: \$730,000

So, the net benefit accumulates over the years, reaching \$270,000 by the end of Year 2 and \$500,000 by the end of Year 3, and finally \$730,000 by the end of Year 4. This calculation shows the progressive accumulation of savings and benefits from the automated invoice processing system.

7. Impact Analysis

Impact Analysis using POPIT model will help identify how change will affect the following areas: (People, Organization, Processes, Information, and Technology)

People: If a new automated system is implemented, there may be some job loss or job transitions. The initial morale will be low. The new automated system will require training and adapting to a new way to processing invoices. The positive side is the automated system will help to eliminate many of the pain points the employees were experiencing (quicker invoice processing, lower error rates, on-time payment rate to suppliers)

Organization: The initial cost to the company will be high. They have to prepare the budget for this expenditure. The organization has to prepare for potential backlash due to job cuts. The positive is the organization will be able to see greater efficiency in the accounts payable department. The invoice automation will free up employees to work

other pressing matters, like paying suppliers. The automated system will also help the organization save money. They will be able to use the savings in other areas of their business.

Processes: Automation will allow staff to concentrate on other area of AP that is still causing problems, like vendor payments and invoice approvals. Additional tools like lean management can be implemented in this area to improve workflow and reduce redundancies. Communication channels could be improved with vendors and departments to ensure invoices are paid on time.

Information: Automation may help standardize data so that all the information is aligned and allow for data analysis ease. Automation will also create more accurate database which will allow for effective decision making. Fully integrating with ERP system will ensure the data will be captured in existing database which will ease the burden of having data in different places.

Technology: The automation requires a large startup cost and will require monthly fees to use the existing technology. As invoice volume scale up, the monthly fee may also scale. Maintenance and troubleshooting will be necessary to keep system running smoothly. Additional hardware like servers and storage may be needed to support automation system.

8. Risk Analysis

The key risks identified include resistance to change, data migration issues, and training and adoption issues, which have high risk levels and require robust mitigation strategies.

Other risks such as integration challenges, initial costs, and system reliability have been assessed as medium but still need careful management to ensure successful implementation.

By addressing these risks proactively, Northwind Enterprises can enhance the likelihood of a smooth transition to the new automated invoice processing system.

Resistance to Change

- Description: Employees might resist the new automated system due to fear of job loss or discomfort with new technology.
- Impact: High
- Likelihood: High
- Proposed mitigation strategies: Engage employees early, provide clear communication about the benefits, offer extensive training, and explore reassignment opportunities for affected staff.

Data Migration Issues

- Description: Potential errors or loss of data during the migration from the manual system to the automated system.
- Impact: High
- Likelihood: Medium
- Proposed mitigation strategies: Develop a detailed data migration plan, conduct thorough testing, and validate data post-migration.

Integration Challenges

- Description: Technical difficulties in integrating the new system with the existing ERP system.
- Impact: Medium
- Likelihood: Medium
- Proposed mitigation strategies: Close collaboration with IT and vendors, allocate resources for customization and testing, and ensure thorough testing before full deployment.

Initial Costs and Budget Overruns

- Description: The project might exceed the budget due to unforeseen customization and integration costs.
- Impact: Medium
- Likelihood: Medium
- Proposed mitigation strategies: Establish a contingency budget, closely monitor project costs, and adjust plans as necessary to stay within budget.

Training and Adoption Issues

- Description: Employees might struggle to learn and adopt the new system effectively.
- Impact: Medium
- Likelihood: High

- Proposed mitigation strategies: Provide comprehensive and ongoing training, offer support resources, and create a feedback loop to address issues promptly.

System Downtime and Reliability

- Description: The new system may experience downtime or reliability issues, impacting operations.
- Impact: High
- Likelihood: Low
- Mitigation: Choose a reliable vendor, ensure robust support and maintenance agreements, and have backup processes in place.

RISK MATRIX

Risk	Impact	Likelihood	Risk Level (Impact x Likelihood)	Mitigation
Resistance to Change	High	High	Very High (9)	Engage employees, provide training, and explore reassignment opportunities.
Data Migration Issues	High	Medium	High (6)	Develop a detailed migration plan, conduct thorough testing, and validate data post-migration.
Integration Challenges	Medium	Medium	Medium (4)	Collaborate with IT and vendors, allocate resources for customization, and ensure thorough testing.
Initial Costs and Budget Overruns	Medium	Medium	Medium (4)	Establish a contingency budget, closely monitor costs, and adjust plans as necessary.
Training and Adoption Issues	Medium	High	High (6)	Provide comprehensive training, offer support resources, and create a feedback loop.
System Downtime and Reliability	High	Low	Medium (3)	Choose a reliable vendor, ensure robust support agreements, and have backup processes in place.

9. Implementation Plan

Phase 1: Requirements Gathering and System Design (0-2 months)

- Conduct workshops and interviews with key stakeholders to understand detailed business requirements.
- Document current processes and pain points.
- Define the scope and objectives of the new system.
- Design the system architecture and workflow.

Phase 2: System Development and Customization (2-4 months)

- Develop the core functionalities of the automated invoice processing system.
- Customize the OCR and validation algorithms to meet specific business needs.
- Build integration modules to connect with the existing ERP system.

Phase 3: System Integration and Testing (4-5 months)

- Integrate the automated system with the ERP and other relevant systems.
- Conduct thorough testing, including unit testing, integration testing, and user acceptance testing (UAT).
- Validate data migration and ensure data integrity.

Phase 4: Training and Deployment (5-6 months)

- Develop comprehensive training materials and conduct training sessions for all relevant staff.
- Deploy the system in a phased manner to minimize disruption.
- Provide ongoing support and address any issues that arise during the initial deployment phase.

10. Project Monitoring & Evaluation

To ensure the successful implementation of the automated invoice processing system at Northwind Enterprises, a robust framework for project monitoring and evaluation will be established. This framework will focus on continuous project tracking and the assessment of key performance indicators (KPIs) to measure the project's success.

Project tracking will involve regular progress reviews, milestone assessments, and issue resolution mechanisms to keep the project on schedule and within budget.

Weekly status meetings with the project team, including IT staff, finance personnel, and external vendors, will be conducted to review progress, identify any potential risks or bottlenecks, and implement necessary corrective actions.

Detailed progress reports will be generated bi-weekly, highlighting completed tasks, ongoing activities, and any deviations from the project plan.

Key Performance Indicators (KPIs)

The evaluation of the project's success will be based on specific KPIs that align with the project's objectives. Key KPIs will include:

- error rates in invoice processing,
- processing time per invoice,
- labor costs,
- system uptime.

A successful project implementation will see a significant reduction in invoice processing errors and time, directly translating to increased efficiency and cost savings.

The reduction in labor costs will be tracked by comparing pre- and post-implementation staffing levels and associated wages. Additionally, user satisfaction will be monitored through surveys and feedback sessions, ensuring that the system meets the needs of the Accounts Payable (AP) department and other stakeholders.

Regular performance evaluations will be conducted at key project milestones (3 months, 6 months, and 12 months post-implementation) to measure these KPIs and make any necessary adjustments to optimize the system's performance.

This structured approach to monitoring and evaluation will ensure the project's objectives are met and sustained over the long term.

11. Conclusion

Northwind Enterprises currently faces significant challenges with its manual invoice processing system, including high error rates, time-consuming procedures, and scalability issues. After considering three potential solutions—doing nothing, hiring additional staff, and implementing an automated invoice processing system—the analysis strongly supports the implementation of an automated system. This proposed solution includes Optical Character Recognition (OCR) for data capture, real-time invoice validation against purchase orders, and seamless integration with the existing Enterprise Resource Planning (ERP) system.

The financial analysis reveals substantial benefits, with projected annual labor cost savings of \$290,000 and net annual savings of \$230,000 after accounting for ongoing licensing fees. The initial investment of \$190,000 for customization and integration is expected to be recouped within the first year, with a Return on Investment (ROI) of 384.21% over four years. The net benefit accumulates to \$730,000 by the end of the fourth year, demonstrating a highly favorable financial outcome.

A comprehensive capability assessment shows that while Northwind Enterprises has a solid foundation in its current IT and human resources, enhancements in training, change management, and system integration are necessary. The proposed project monitoring and evaluation framework includes regular progress reviews, milestone assessments, and key performance indicators (KPIs) to ensure the project's success and sustainability.

Alternative solutions, such as maintaining the status quo or increasing staffing levels, do not address the fundamental inefficiencies and result in higher long-term costs. Implementing the automated system not only reduces labor costs and errors but also enhances scalability, real-time tracking, and overall operational efficiency.

In conclusion, the implementation of an automated invoice processing system is the most viable and beneficial solution for Northwind Enterprises. It addresses current pain points, offers significant cost savings, and positions the company for future growth and competitiveness. With careful planning, robust training, and thorough monitoring, Northwind Enterprises can achieve a seamless transition to an automated system and realize substantial long-term benefits.